

13.3 Comparing Key Features of Functions

Lesson Introduction

 Benchmarks	MA.912.F.1.6 Compare key features of linear and nonlinear functions, each represented algebraically, graphically, in tables, or as written descriptions.		 Learning Targets	- I can compare key features of linear and nonlinear functions that are represented algebraically, graphically, in tables, or in written descriptions. - I can verify when a quantity increasing exponentially exceeds a quantity increasing linearly or quadratically.
	MA.912.F.1.8 Determine whether a linear, quadratic, or exponential function best models a given real-world situation.			
 Benchmark Coherence	Building On MA.8.F.1.2 Given a function defined by a graph or an equation, determine whether the function is a linear function. Given an input-output table, determine whether it could represent a linear function.		Working Toward MA.912.F.1.7 Compare key features of two functions, each represented algebraically, graphically, in tables, or as written descriptions.	
 Essential Questions	- How can I compare key features of linear and nonlinear functions expressed algebraically? - How can I compare key features of linear and nonlinear functions expressed graphically? - How can I compare key features of linear and nonlinear functions expressed in tables? - How can I compare key features of linear and nonlinear functions expressed in a written description?			
 Lesson Narrative	In this lesson, students utilize their prior learning in the course to compare functions of various types. Students are presented with different types of functions presented in various ways to build upon what they have learned in prior units. Students begin by comparing functions using different key features of the functions displayed in multiple representations. They then transition into real-world application problems using the various functions and key features.			
 Component Summary	13.3.1 Warm-Up (~5 min.)	Pattern analysis activity (SE p. 247)	13.3.3 Collaboration (20 min.)	Real-world application collaboration activity (SE p. 250)
	13.3.2 Guided Instruction (20 min.)	Instruction activity with comparing functions in different representations (SE p. 247)	13.3.4 Wrap-Up (~5 min.)	Self-Reflection activity (SE p. 253)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		Calculators	
			Additional Preparation	
			N/A	

 Teacher Tips	Common Misconceptions • Students may not recall how to identify specific key features for a given function. • Students may misrepresent data in a specific form (i.e., table, graph, or equation). • Students may be unable to identify a function type when expressed in a specific form.	Things to Consider • If time is an issue, consider having students complete 13.3.3 Collaboration for homework. • Consider having students provide answers to domain and range using both inequalities and in set-builder notation for additional practice with each notation. • Consider working with a small group of students who are struggling during 13.3.3 Collaboration.
	Literacy and Language Routines Three Reads 13.3.3 Collaboration provides students the opportunity to collaborate on real-world applications of various function types. For this activity, consider leveraging the Three Reads routine. This routine supports reading comprehension of problems and meta-awareness of mathematical language.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.4.1: Discussion and Reflection 13.3.3 Collaboration provides students a collaboration activity with real-world applications of functions. During this process, students will be communicating mathematical ideas, vocabulary, and methods effectively and compare the efficiency of their methods to those expressed by their partners.
	 Differentiate Instruction	This lesson provides different function types in the problems provided in this lesson. It may be a challenge to complete all of them because of the richness of each problem. Consider working through each problem and timing yourself, then multiplying by 2.5 to get an estimate of how long students might take to complete. Use data from the associated prior units to determine which problems you want to do, and which ones may need to be omitted. Also, consider differentiating the problems into student groups based on their individual strengths and weaknesses.
Lesson Notes:		

13.3.1 Warm-Up: Tale of Three Patterns

Component Overview: Students analyze three different patterns to determine how many squares will be in the fifth step.



13.3.1 Warm-Up: Tale of Three Patterns

- The table shows three different patterns.

Step	Pattern A	Pattern B	Pattern C
1			
2			
3			
4			
5			



Consider leveraging a Think-Pair-Share routine with this activity. Allow students a couple minutes to try this activity on their own and then, when time is called, work with a partner to review answers and their strategies for approaching the patterns. Allow an opportunity for students to work with other pairs or have a whole-group discussion to discuss strategies and approaches.

- Write the number of squares needed for step 5 of each pattern.

Pattern A: 26 squares

Pattern B: 13 squares

Pattern C: 32 squares

- Which pattern increases the same number of squares for each step?

Pattern B

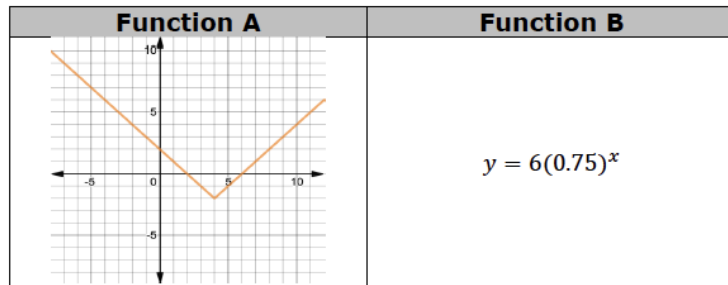
13.3.2 Guided Instruction: Comparing Functions

Component Overview: Students will work to compare functions presented in different forms.



13.3.2 Guided Instruction: Comparing Functions

1. The table shows two functions.



- Which function has the smallest y -intercept?
 - ☐ **Function A**
 - ☐ Function B
- As $x \rightarrow \infty$, which function has larger values of y ?
 - ☐ **Function A**
 - ☐ Function B
- Compare the ranges of the functions and indicate any similarities or differences.
Range of Function A is $y \geq -2$ and Function B is $y > 0$. While both function ranges continue toward positive infinity, Function A has negative range values while Function B does not.
- On the interval where Function A is negative, what type of values does Function B have?
The types of values are positive, which are above the x -axis.



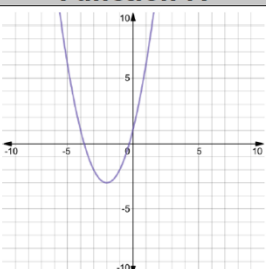
Consider the use of a variety of problems and spiraling to help students make connections with the content they have learned. Teachers may determine alternate methods to teaching this activity based on your students' needs.



What are the two function types presented in Function A and B?
How do you approach comparing their key features?

13.3.2 Guided Instruction: Comparing Functions (continued)

2. The table shows three different functions.

Function H	Function K	Function M
$y = 5(1.35)^x$	x	
	y	
	-2	
	-1	
	0	
	1	
	2	



Consider a differentiation opportunity if you will not be using all the problems. Consider customizing each question as a “set” and providing students groups with a set based on their needs from the data from the corresponding units that the function types are from. Students can complete multiple sets in a given time frame.

a. Determine the type of function that each represents each

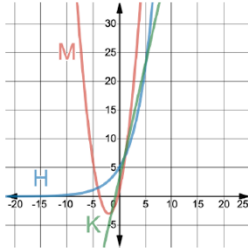
Function H	Function K	Function M
☐ linear ☐ quadratic ☒ exponential	☒ linear ☐ quadratic ☐ exponential	☐ linear ☒ quadratic ☐ exponential

b. Which key features are the same for all three functions?

- ☐ domain
- ☐ range
- ☐ x -intercepts
- ☐ y -intercepts
- ☐ right end behavior
- ☐ left end behavior
- ☐ intervals where the function is positive
- ☐ intervals where the function is negative
- ☐ intervals where the function is increasing
- ☐ intervals where the function is decreasing

13.3.2 Guided Instruction: Comparing Functions (continued)

3. The graph shows **Function H**, **Function K**, and **Function M**.



Which function, **H**, **K**, or **M**, will eventually exceed the other functions as all the functions increase?

- ☐ **Function H**
- ☐ **Function K**
- ☐ **Function M**

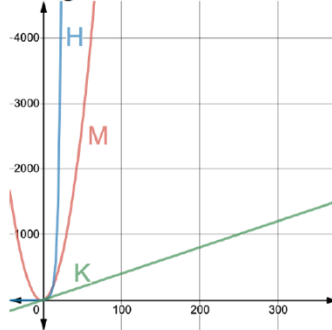


What do you think it means to eventually exceed the other functions?

How does that look on the graph?

What are the three types of functions presented?

4. The graph shows **Function H**, **Function K**, and **Function M** using different scales for the axes.



Discuss with your partner whether or not the graph verifies that the function you chose in question 3 will eventually exceed the other functions as all the functions increase. If necessary, adjust your answer.

Yes, function H is exponential which means your results are rising higher than the other ones.

13.3.2 Guided Instruction: Comparing Functions (continued)

5. Comparing simple interest to compound interest is like comparing a linear function to an exponential function. As observed in the previous lesson, the amount of interest accrued with compound interest exceeds simple interest. In the graphs of functions **H**, **K**, and **M**, the quadratic function, **M**, rises faster than the linear function, **K**, but it does not rise faster than the exponential function **H**.

Discuss with your partner why exponential growth will always eventually exceed a parabola that opens up, and a line with a positive slope. Summarize your discussion.

Answers will vary. Exponential is based on multiples or high exponents which is a greater increase. Linear is increasing consistently by adding values. Quadratics are increasing by only the second power.



If you differentiate the prior four questions, it is important for all students to complete question 5. This question is rooted in the learning students have done in the last two lessons of this current unit. This will provide students additional spiraled practice from the current unit.

13.3.3 Collaboration: Digital Music Sales

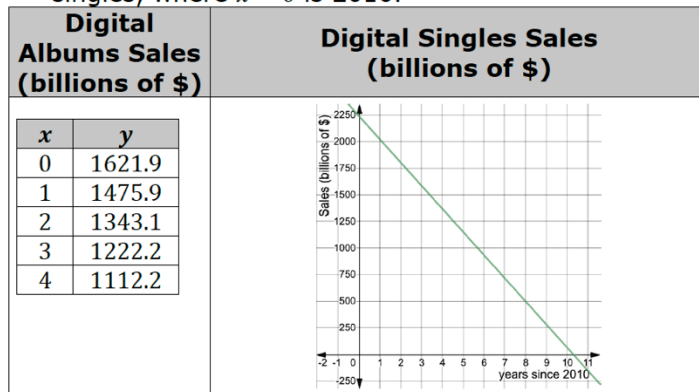
Component Overview: Students will complete two questions with a partner or small group on real-world applications of functions.



13.3.3 Collaboration: Digital Music Sales

Work with your partner to complete the following.

- The table shows a table of values representing the revenue, in billions of dollars, from sales of digital albums, where $x = 0$ is 2010, and a graph representing the revenue, in billions of dollars, from sales of digital singles, where $x = 0$ is 2010.



- Approximate the x -intercept for the function that represents sales of digital singles.
 ≈ 10.25
- Interpret the x -intercept of the function that represents sales of digital singles.
Approximately a little over 10 years after 2010 there were no sales profits for digital singles.
- Discuss with your partner whether or not the interpretation is accurate. Summarize your discussion.
Answers will vary. Yes, this could be accurate and the graph slowly decreased over time.
- What is the domain of the function representing sales of digital singles?
 $0 \leq x \leq 10.25$



For this activity, consider using the Three Reads routine. This routine supports reading comprehension of problems and meta-awareness of mathematical language. If needed, consider modeling for students this routine with the first question if necessary.



This Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. Students may revise their thinking throughout the Exploration as they engage with each new question.

As students have worked with these types of functions individually, comparing the features of the functions offers students the chance to utilize spaced recall. Encourage students who do not remember something to utilize their resources. By engaging in a productive struggle to remember, students are more likely to remember the content.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.



What information from the question or graph can help you interpret the x -intercept?

13.3.3 Collaboration: Digital Music Sales (continued)

- e. Work with your partner to determine the key features for each function.

Key Feature	Digital Album Sales	Digital Single Sales
y-intercept	\$1,621.90	≈ \$2,250
End Behavior	As $x \rightarrow -\infty$, $y \rightarrow 1621.9$	As $x \rightarrow -\infty$, $y \rightarrow 2250$
	As $x \rightarrow \infty$, $y \rightarrow 0$	As $x \rightarrow \infty$, $y \rightarrow 0$
Increasing or Decreasing	The function is <u>decreasing</u> over time.	The function is <u>decreasing</u> over time.



What did you consider to be the most helpful in determining what type of function was being represented?

- f. Discuss with your partner what type of function best describes the table of values that represents the digital album sales. Explain your choice.
- linear
 - quadratic
 - **exponential**

The x -values are constantly increasing, while the y -values are decreasing at a constant percent rate of change, 0.91.

2. The table shows written descriptions and equations that model the home prices, in thousands of dollars, for two different time periods.

Home Prices in Florida	
2009-2014	2015-2019
The average price for a house in Florida between 2009 and 2014, y , in thousands of dollars, is modeled by the equation $y = 4.4x^2 - 22.1x + 160.8$, where x is the number of years since 2009.	In 2015, the average home price in Florida was 175.6 thousand dollars. For each year between 2015 and 2019, the average price increased by 19.1 thousand dollars.

13.3.3 Collaboration: Digital Music Sales (continued)

- a. Work with your partner to determine the key features. Give the domain in set-builder notation.

Key Feature	Home Prices in Florida 2009-2014	Home Prices in Florida 2015-2019
Domain	$\{x: 0 \leq x \leq 5\}$	$\{x: 0 \leq x \leq 4\}$
y-intercept	160.8	175.6



Some students may still be struggling with set-builder notation. If students have the correct values but do not have the correct notation, this means the students need additional support with set-builder notation, not determining the domain.

- b. Which context has the larger y-intercept?
Home prices in Florida in 2015 – 2019.
- c. From standard form, the vertex of the parabola can be found using $\left(\frac{-b}{2a}, c - \frac{b^2}{4a}\right)$. Determine the vertex of the parabola that models home prices in Florida from 2009 to 2014.
(2.5, 133.05)
- d. Use the domain and the vertex to determine the range for home prices in Florida from 2009 to 2014.
 $133.05 \leq y \leq 160.8$
- e. Determine the type of function that models the home prices for 2015 to 2019.
linear
- f. Use the information given for the home prices for 2015 to 2019 to determine the range of prices.
 $175.6 \leq y \leq 252$
- g. Which period, 2009–2014 or 2015–2019, has the greatest range of prices?
2015 – 2019, the linear situation can have larger growth in the earlier periods of the situation.

13.3.4 Wrap-Up: Reflection

Component Overview: Students will complete a self-reflection on what they knew, what they wondered, and what they learned in this lesson.



13.3.4 Wrap-Up: Reflection

1. Complete the table.
Answers will vary.

The terms and information used in the Guided Instruction and Collaboration that I knew were ...	Something I wondered during the Guided Instruction or Collaboration was ...	Something I learned during the Guided Instruction or Collaboration was ...



The information provided from the students here, coupled with what was observed during instruction and collaboration, can provide information for how engaged each student was during the lesson. This provides an individual accountability of a student's engagement (MA.K12.MTR.1.1). Consider reviewing student responses to consider which students may need further support in effortful learning.